COMMERCIAL ITEM DESCRIPTION

SUCTION APPARATUS, OROPHARYNGEAL

Tracheal, Aeromedical, Portable/Dual Voltage

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a portable, battery operated, oropharyngeal suction apparatus, 120 volts, 50/60/400 Hz, AC, with separate stepdown transformer for operation on 230 volts, 50/60 Hz, AC.

Salient characteristics:

Suction apparatus. The suction apparatus shall be a portable, oropharyngeal tracheal suction apparatus. It shall be self-contained in a carrying case complete with a rechargeable battery, suitable for use in hospitals, ambulances, and field uses with no external power available, and in military aeromedical aircrafts. It shall be suitable for providing aspiration of secretions, blood or vomitus during emergency treatment of an unconscious or injured patient.

Major components. The suction apparatus shall be provisioned with the following integrated components: vacuum pump and motor, battery charger, internal battery, interlocking switch(es), indicating lamps, circuit breaker, EMI and RFI shielding circuitry, collection container, patient suction tube, tracheal catheters, AC power cord, vehicle power cord, vacuum valve, vacuum gauge, rinse bottle, litter mounting attachment, and separate transformer.

Vacuum pump and motor. The design of the pump-motor shall be such that overflow of patient debris shall not jam or clog it during usage. It shall be readily cleanable for next use.

Vacuum range. The vacuum capacity shall include an adjustable range from 0 to at least 500 mmHg. The high end of this range shall be adjustable by a control valve which regulates the pump output vacuum. Using the internal battery, the unit shall be capable of maintaining the specified vacuum for at

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least 30 minutes. This requirement shall be tested in accordance with ASTM F960 using the internal battery, the 120 amp hour external battery, and using 110 volts AC at 50, 60, and 400 Hz. It shall also be tested using its internal battery in accordance with test C, "Vacuum Capacity Using Battery" stated herein.

Flow rate. The free airflow at the patient end of the suction tube shall be at least 30 LPM using any of the specified power sources (see paragraph titled "Power Sources") and operating at specified temperature extremes (see paragraph titled "Operating Temperature Range"). This requirement shall be tested in accordance with ASTM F960 using the internal battery, the 120 amp hour external battery, and 110 volts at 50, 60 and 400 Hz.

Battery charger. This apparatus shall be provisioned with a battery charger, operational from nominal 110-125 volts, 50/60 Hz. It shall be capable of recharging the totally depleted internal battery within 24 hours.

Internal battery. The apparatus shall be equipped with a 12 volt internal battery capable of operating the unit in accordance with the requirements specified herein. (Multiple batteries wired to provide 12 volts DC is acceptable).

Totally depleted internal battery shall be rechargeable within 24 hours. This requirement shall be tested in accordance with test B, "Recharge Time" stated herein.

Internal battery operating time. The battery operating time, with vacuum adjusted at 120 mmHg, shall be at least 30 minutes when cycled through 5 cycles of 6 minutes each. Each cycle shall have periods of a 2 minute period with patient tube occluded, a 1 minute period with patient tube open (free air flow), a 2 minute period with patient tube occluded, and a 1 minute period with the pump turned off (rest period). This requirement shall be tested in accordance with test C, "Fully Charged Battery Operating Time" stated herein.

Interlocking switch(es). The apparatus shall be provisioned with interlocking switch(es) to select "OFF/CHARGE", "AC/ON", "EXT. DC/ON", and "INT. BAT/ON". Switch arrangement shall not allow inappropriate selection of circuitry, causing damage to apparatus.

Indicating lamps. The apparatus shall contain separate indicating lamps, indicating "PUMP ON" and "CHARGER ON".

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Electromagnetic interference (EMI) and radio frequency interference (RFI). The apparatus shall be designed to control electromagnetic interference (EMI) and radio frequency interference (RFI) aircraft (including associated ground support equipment and portable medical equipment used for aeromedical airlift). Proof of compliance shall be accomplished by aeromedical testing of first article sample dedicated solely for these tests.

Collection container. The collection container shall have a minimum capacity of 500 ML. The container shall meet the testing requirements of ASTM F960.

Patient suction tube. The apparatus shall be provisioned with a patient suction tube not less than 45 inches long.

Tracheal suction catheters. The apparatus shall be provisioned with 2 catheters, both with thumb vacuum control to adjust vacuum level. Two sizes are required, one 14 FR and the other 18 FR.

AC power cord. The apparatus shall be provisioned with a power cord at least 6 feet long, with hospital grade male plug.

Vehicle power cord. The apparatus shall include a 6 foot electric cord assembly to obtain power from external vehicle DC source.

Vacuum valve. The apparatus shall be provisioned with a control valve (in addition to catheter thumb control), which shall regulate the high end of vacuum level.

Vacuum gauge. The apparatus shall be provisioned with a vacuum gauge calibrated both in millimeters and inches of mercury. It shall include the range of 0 to 500 mmHg .

Rinse bottle. A rinse bottle shall be provided with at least 250 ml capacity.

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Litter mounting attachment. The self-contained apparatus shall include a litter mounting attachment consisting of four D-rings, two attached to the case front and two attached to the case rear, and two 24 inch long hook and pile/nylon webbing attachment straps. The straps shall be shipped in the carrying case.

Separate stepdown transformer. The unit shall include a separate stepdown transformer as a separate component not required to be in the carrying case, but shipped in the same shipping package. The transformer shall accept 220-250 volts, 50/60 Hz and provide one half output voltage for the suction apparatus. The transformer shall include a hospital grade, 110 volt receptacle to accept the suction apparatus' specified male line cord. The transformer shall include 3 posts in a compartment with a wire strain-relief attachment for SJT-16/3 wire. The three posts shall be wired to accept any worldwide wiring scheme required to connect a cord for 220-250 volts, single phase AC service. The separate stepdown transformer shall be rated for continuous standby use. The transformer shall comply with UL-1012, including testing.

Power sources. The self-contained apparatus shall be operational from external 12-14 volts DC vehicle generating system rated at 120 amperes at 14 volts, external 120 amp-hour heavy duty 12 volt battery, 12 volt internal battery, and 110-125 volt, 50/60/400 Hz power sources.

In addition, the separate transformer shall be employed as a power source for the apparatus by reducing 220-250 volts, 50/60 Hz to 110-125 volts.

Operating temperature range. The operating temperature range shall include $-4^{o}F$ to $+115^{o}F$. The apparatus shall meet the requirements in accordance with ASTM F960 using the internal battery, the 120 amp hour external battery, and 110 volts AC at 50, 60, and 400 Hz.

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AC operating time limitation. The apparatus shall be rated to operate 27 minutes per hour with a 10 year life when in AC operation run mode. For this purpose a 10 year life is defined as being in AC run mode for 27 minutes per hour, 5 hours per day, 3 days per week, 50 weeks per year, for 10 years. (Continuous 24 hour charging mode is allowed.) Circuit breaker(s)/fuse(s) shall protect the pump and transformer(s) of the apparatus in case of current overload. A warning label shall be placed on the unit such as "RECOMMENDED AC USAGE SHOULD NOT EXCEED 27 MINUTES PER HOUR".

DC operating time. The apparatus shall be rated for continuous duty on external DC battery operation.

Cleaning and sterilizing. The catheters, patient tubing, and plastic collection container shall be sterilizable via ethylene oxide gas and cold liquid disinfectants. They shall be readily removable without tools for emptying and cleaning during emergencies.

Carrying case. The carrying case shall be fabricated of semi-rigid, dual-wall, high density suitable plastic, or of corrosion-resistant metal. It shall house all components except possibly the transformer and/or the two specified sets of operating and maintenance instruction manuals. Components not in the carrying case shall be in the unit package.

Brief operating instructions. The apparatus shall be supplied with brief operating instructions which shall be markings on the case. The brief instructions are in addition to the specified operating and maintenance manuals.

Maintainability. The apparatus shall be capable of being disassembled and assembled for maintenance without use of special tools.

Weight. The apparatus in its carrying case, less the transformer and the two sets of specified operating and maintenance instruction manuals, shall not weigh more than 20 pounds.

Dimensions. Dimensions of the carrying case shall not exceed 20 inches by 15 inches by 12 inches high.

Marking. The carrying case shall be marked with commercial identification and the National Stock Number (NSN).

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Workmanship. The complete suction apparatus, carrying case, and separate transformer shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One suction apparatus, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 4.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one suction apparatus as specified in this description.

Sampling for tests. Sampling for tests including dimensional test shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one suction apparatus.

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First article. Two first article samples are required, one dedicated solely for EMI and RFI testing by biomedical personnel. All other required tests shall be performed on the second sample.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The following tests shall be conducted to determine compliance with this specification.

a. Minimum suction test. Using a Hg column with the apparatus running at room temperature, measure at least 500 mmHg suction at patient end of the suction tube.

b. Recharge time test. At room temperature, operating on the internal battery, run the apparatus until the internal battery(s) has expired. Condition the battery at -4°F for 8 hours. Next, using any one of the specified power supplies while temperature is rising from -4°F to 60°F during the first 1/2 hour then maintaining 60°F, charge the battery using the component charger, for a period of time not to exceed 24 hours. Test the battery for having a full charge in accordance with the instructions of the battery manufacturer. This test shall be conducted once per specified power supply, testing the charging time using each of the specified power sources (except internal battery and 110 volts, 400 Hz).

c. Fully charged battery operating time, operating temperature range, free airflow at patient end of suction tube. Using the internal battery that was charged in test "b" above (the last test conducted in b), without additional charge, condition the apparatus at 60°F for 8 hours. Next, in ambient temperature of -4°F run the apparatus for at least 35 minutes. During the first 32 minutes, alternately occlude and open (to atmosphere) the suction tube for 2 minute periods. Measure a free airflow of at least 30 LPM at the patient end of the suction tube during the last 2 minutes. Measure the vacuum to be at least 500 mmHg during the last minute (thumb valve occluded). Repeat this test except leave the patient tube continuously open to atmosphere during the first 34 minutes. Measure the free airflow to be at least 30 LPM during the 34th minute. Then measure the vacuum to be at least 500 mmHg during the last minute.

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d. Repeat of test c. Repeat test c once for each electrical power source specified in the General Requirements except use the power sources specified in the General Requirements in lieu of internal buttery, and run the tests in ambient temperature of 115°F. Repeat test c two more times, once using external 12 VDC power supply and once using external 14 VDC power supply in -4°F environment. Condition the test unit at room temperature (about 72°F) prior to each of these tests for at least 8 hours.

e. Collection container capacity test. Measure the volume of water that the collection container is capable of containing to be at least 500 ml.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4617

Location: CID/SUCTION/APPAR/
OROPHARYNGEAL/H1D-35

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54039	SUCTION APPARATUS, OROPHARYNGEAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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